YELLAPANTHULA SAI SAMPATH

ysampath36@gmail.com | +91 70135 12100 | linkedin.com/in/just-sampath | just-sampath.in

PROFESSIONAL SUMMARY

Full Stack & AI Engineer with **3+ years of experience building production RAG systems, enterprise chatbots, and agentic workflows**. Reduced RAG response time from **60~80 seconds to 10–15 seconds at p99**, delivered Slack and Microsoft Teams integrations, and moved seven weekly WFM reports from manual processes to AI-assisted workflows. Experience includes AWS Lambda and Step Functions, Azure Functions, vector databases, and hybrid search.

TECHNICAL SKILLS

AI & LLM: RAG architecture, prompt engineering, RAG evaluation, OpenAI API, Claude, Claude Code, LlamaIndex, AutoGen (pyautogen), Portkey, Langchain, LangGraph, Langfuse
Languages: Python, TypeScript, JavaScript
Retrieval & Vector Search: Pinecone, Qdrant, ChromaDB, BM25, BM42, HNSW index optimization, Jina, Cohere
Cloud & DevOps: AWS Lambda, Step Functions, S3, SQS, API Gateway, ECS/Fargate, ECR, Azure Functions, Docker, CI/CD, Git
Backend & Integrations: Node.js, Express.js, FastAPI, REST APIs, webhooks, BullMQ, Slack Bolt, Microsoft Teams Graph API
Databases & Caching: MongoDB, MySQL, Postgres, Redis, Valkey
Frontend: React, HTML, CSS

PROFESSIONAL EXPERIENCE

Software Engineer II

January 2026 - Present

Concentrix

- Optimized RAG chatbot workflows **with parent-child chunking, model routing, and parallel retrieval**, **reducing response latency and improving answer quality** for production banking use cases.
- **Delivered a modular banking RAG chatbot to production in 3 days** using hybrid search, hierarchical chunking, and evaluation checkpoints, enabling faster deployment of an enterprise AI assistant.
- Built a **Claude Code plugin with 12 agents, 7 hooks, 9 verification gates, 16 skills, 4 commands, and reusable templates**, enabling rapid development of end-to-end report automations from source artifacts.
- Developed **WFM reporting pipelines with Claude**, handling query generation, data aggregation, and production deployment on **Azure Functions**.
- Supported a fast-paced delivery cadence, converting **7 manual WFM reports per week into AI-assisted workflows** across query and aggregation-heavy use cases.

Team Lead

May 2025 - December 2025

Chat Aid | Remote

- **Led cross-functional development across Node.js, Python RAG services, and React**, driving technical decisions and delivery across backend and frontend systems.
- Architected and owned a **serverless document processing pipeline using AWS Step Functions with 7 specialized Lambda functions**, enabling parallel processing and intelligent chunking strategies.
- Reduced RAG response time from **60~80 seconds to 10–15 seconds at p99** by routing models by plan tier, using structured outputs, and reranking results from multiple providers parallelly.
- Enhanced vector database performance by **introducing tenant-specific indexing, recency bias for search results**, and configurable top K retrieval to **improve RAG relevance and search precision**.
- **Designed tree based chunking strategies** for Notion and Confluence knowledge bases, improving context retention and answer relevance in RAG retrieval.
- Added **post-evaluation metrics** for Auto Answer feature to measure and continuously improve response quality.
- Deployed **zero-downtime Qdrant infrastructure** on EC2/EBS with SQS/S3 write buffering, health-gated cutover, and instance promotion/retirement.

Backend Lead

January 2025 - May 2025

Chat Aid | Remote

- Led backend development for a **RAG-based AI chatbot platform**, ensuring system stability for Slack and Microsoft Teams integrations.
- Built a **complete AWS development environment from scratch**: frontend (S3 + CloudFront + Cloudflare), backend/Python services (ECR + ECS + Fargate + ALB), Redis caching, and SQS queues.
- Configured **AWS infrastructure** including VPC networking, IAM policies, internal/external ALBs, API Gateway, Route 53 DNS management, and Secrets Manager.
- Redesigned webhook ingestion to **S3 multipart uploads with 15MB batch processing**, enabling scalable handling of large data volumes in Node.js.
- Executed **streaming byte-level file processing** in Python preprocessing pipeline, eliminating memory constraints for **large file ingestion into Qdrant** vector database.
- Established **CI/CD pipelines** deploying containerized applications to ECR, ECS Fargate, and CloudFront CDN; introduced **Chainguard images** for container security.
- Built **customizable answer personas** allowing users to tailor response structure and tone, along with **analytics dashboards** for plans, integrations, and auto-answer performance.

Software Development Engineer I

July 2023 - December 2024

Springworks | Remote

- Architected backend systems for Albus, an **enterprise RAG chatbot** ingesting data from **50+ sources** including JIRA, Notion, Confluence, Google Drive, and custom web crawlers.
- Increased test coverage to **90%** using Mocha and Chai, **reducing production defects** through comprehensive unit and integration testing.
- Streamlined MongoDB queries by eliminating expensive **$lookup (join) operations**; improved query routing logic to reduce unnecessary AI/RAG calls and decrease infrastructure costs.
- Engineered **Auto Answer feature** enabling automatic message responses **without human intervention**, increasing customer support efficiency.

- Built **end-to-end Web Crawler** using Apify for scalable enterprise-grade data collection with proper rate limiting and error handling.
- Created **Custom API endpoints** enabling **programmatic RAG access** for enterprise customer integrations and third-party system connectivity.
- Accelerated Slack data ingestion using BullMQ **parent-child queue architecture** with **tenant-channel isolation**, ensuring efficient concurrent processing.
- Led **Actions feature development**: rearchitected agentic system from WebSockets to **webhooks** for reliable JIRA ticket creation, meeting scheduling, and CRM pipeline updates.
- Made AI agent tool calls **deterministic** by shifting responsibility from LLM decision-making to **structured parameter extraction**, significantly improving task completion rates.
- Integrated MongoDB state management for resumable conversations, allowing users to continue complex multi-step workflows after interruptions.
- Built Python backend services using FastAPI and pyautogen for **agentic AI workflows**; hardened Docker configurations with multi-stage builds and health checks.
- Expanded **Microsoft Teams platform support** using Graph API, extending product reach to enterprise customers.
- Delivered **AI call automation system** enabling voice-based customer interactions.

PROJECTS

Journal App | Node.js, Express, MongoDB, Pug

- Created a **full-stack web journal application** with secure user authentication, session management, and complete CRUD operations.
- Applied server-side rendering with Pug templating, input validation middleware, and MongoDB schema design with proper indexing.

Stock Price Prediction Model | Python, scikit-learn, Pandas, NumPy

- Built regression models including decision trees for next-day stock price prediction with feature engineering pipeline.
- Established evaluation framework with train-test splitting, cross-validation, and performance metrics tracking.

EDUCATION

Bachelor of Technology in Computer Science and Engineering (AI/ML)

May 2024

Godavari Global University | *Rajahmundry, India* | *GPA: 8.49/10*

CERTIFICATIONS AND ACHIEVEMENTS

- **Microsoft Certified:** Data Scientist Associate (Python)
- **HackerRank Certifications:** Java, JavaScript, Problem Solving
- **NPTEL Elite + Silver:** Data Structures and Algorithms Using Python (National certification)
- **GeeksforGeeks:** Rank #2 at university